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723.ALLOGENEIC TRANSPLANTATION: LONG-TERM FOLLOW-UP, COMPLICATIONS, AND DISEASE RECURRENCE

Anellovirus in Cerebrospinal Fluid in Patients with Encephalitis and Meningitis after Allo-HSCT: Clinical Characteristics and PrognosisXiujuan Ma¹, Fang Wang, MD², Fang Fang¹, Luyou Han¹, Hongxing Liu, MD^{3,4,1}¹Department of Laboratory Medicine, Hebei Yanda Lu Daopei Hospital, Langfang, China²Department of Clinical Laboratory, Hebei Yanda Lu Daopei Hospital, Langfang, China³Division of Pathology & Laboratory Medicine, Beijing Lu Daopei Hospital, Beijing, China⁴Precision Medicine Center, Beijing Lu Daopei Institute of Hematology, Beijing, China**Background**

Encephalitis and meningitis following allogeneic hematopoietic stem cell transplantation (allo-HSCT) significantly increase patient mortality. Commonly detected pathogens in cerebrospinal fluid (CSF) include herpesviruses and anelloviruses. Anelloviruses are small, circular, single-stranded DNA viruses widely present in various human tissues. However, their clinical impact on patients with encephalitis and meningitis post-allo-HSCT requires further investigation. Metagenomic next-generation sequencing (mNGS) enables unbiased detection of all microorganisms in clinical samples. This study aims to retrospectively analyze the detection rate of anellovirus in CSF and its correlation with clinical events in patients with encephalitis and meningitis post-allo-HSCT.

Methods

We collected CSF mNGS results and clinical data from 152 patients with encephalitis and meningitis post-allo-HSCT at our hospital from 2022 to 2023. Patients were first grouped based on anellovirus positivity: anellovirus-negative (anello-neg) and anellovirus-positive (anello-pos). In the anello-pos group, anellovirus detection results were graded by reads per million (RPM), with RPM ≥ 1 defined as high (anello-high) and RPM < 1 as low (anello-low). Statistical differences in clinical data between groups were analyzed using the Chi-square test and Fisher's exact test. Kaplan-Meier (K-M) curves and multivariate Cox regression analyses were used for survival analysis and to identify risk factors for poor prognosis.

Results

Among 152 patients, 86 were male and 66 were female, aged 3-71 years (median 29 years), 35 cases ≤ 14 years and 117 cases > 14 years. The diagnosis included leukemia (113 cases), aplastic anemia (AA, 17 cases), myelodysplastic syndromes (MDS, 10 cases), and others (12 cases). The primary transplantation type was haploidentical HSCT (haplo-HSCT) in 129 (84.9%) cases. The main pathogens were viruses, with herpesvirus in 97 cases (83.8%) and anellovirus in 84 cases (55.3%). Among the anello-pos patients, 39 were anello-high (46.4%) and 45 were anello-low (53.6%). Other infections included bacterial (4 cases, 2.6%), fungal (8 cases, 5.3%), parasitic (3 cases, 2.9%), and multiple pathogens (13 cases, 8.6%). Median time to anellovirus detection was 127 (31-745) days post-allo-HSCT. Acute graft-versus-host disease (aGVHD) occurred in 88 patients and chronic GVHD (cGVHD) in 56 patients, with no statistical significance in relation to anellovirus detection ($P > 0.05$).

Clinical baseline characteristics between anello-pos and anello-neg groups showed no statistical differences in gender, age, primary disease, donor type, neutrophil engraftment time, platelet engraftment time, aGVHD occurrence, cGVHD occurrence, or other CNS microbial infections ($P > 0.05$). K-M survival analysis showed higher mortality in the anello-high group ($n=39$) compared to anello-low ($n=45$) and anellovirus-neg ($n=68$) groups ($P < 0.05$). No significant difference in mortality was found between anello-low and anello-neg groups ($P > 0.05$). Univariate Cox regression indicated shorter survival time in the anello-high group ($P < 0.05$). Multivariate Cox regression identified anello-high (HR=3.465, 95%CI 1.582-7.589, $P < 0.05$), aGVHD (HR=2.565, 95%CI 1.267-5.191, $P < 0.05$), and intracranial fungal infection (HR=3.543, 95%CI 1.378-9.108, $P < 0.05$) as risk factors for poor prognosis.

Conclusion

In patients with encephalitis or meningitis post-allo-HSCT, high levels of CSF anellovirus are an independent risk factor for poor prognosis. Patients with high CSF anellovirus levels have a poorer prognosis.

Disclosures No relevant conflicts of interest to declare.

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